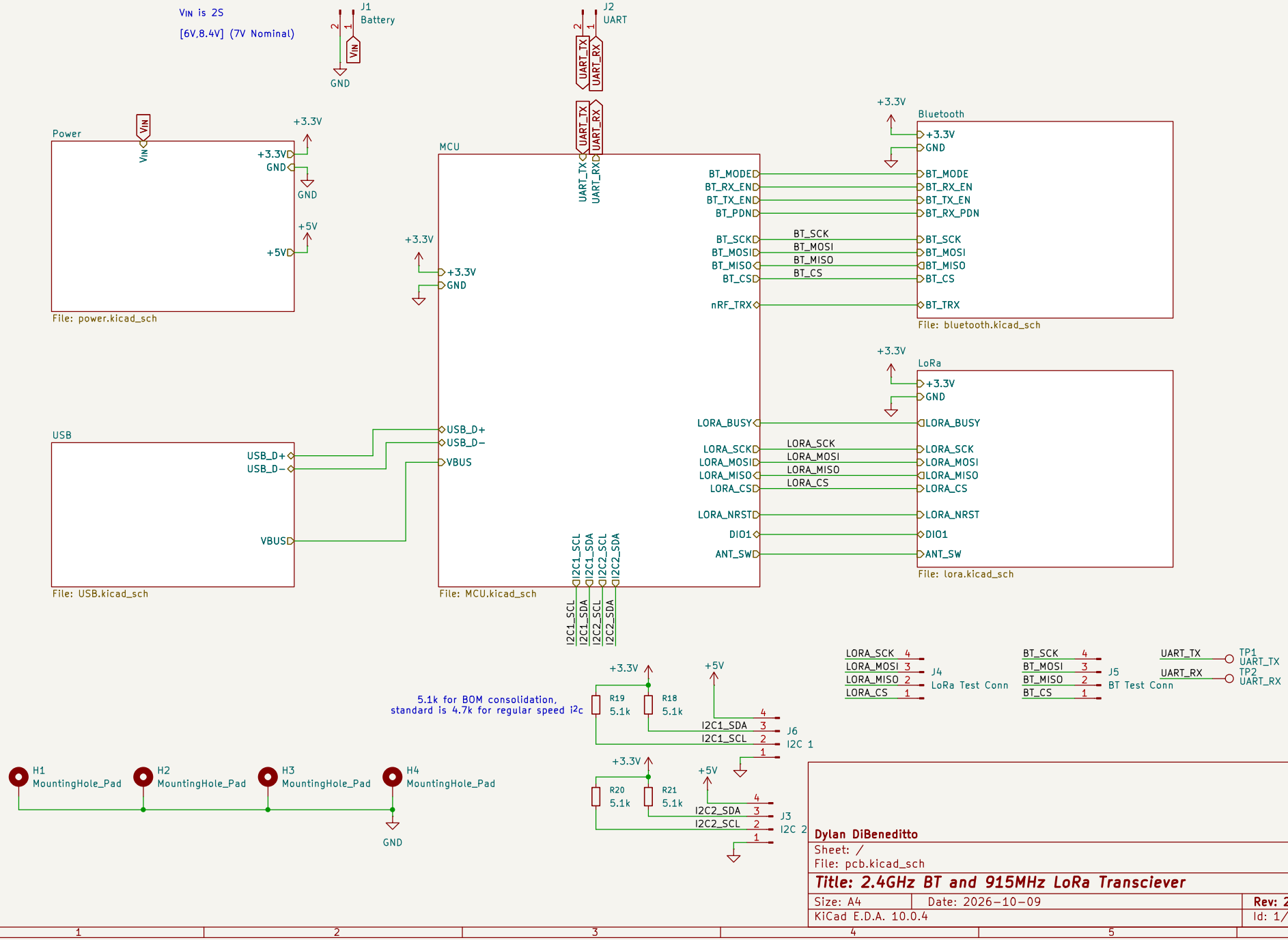
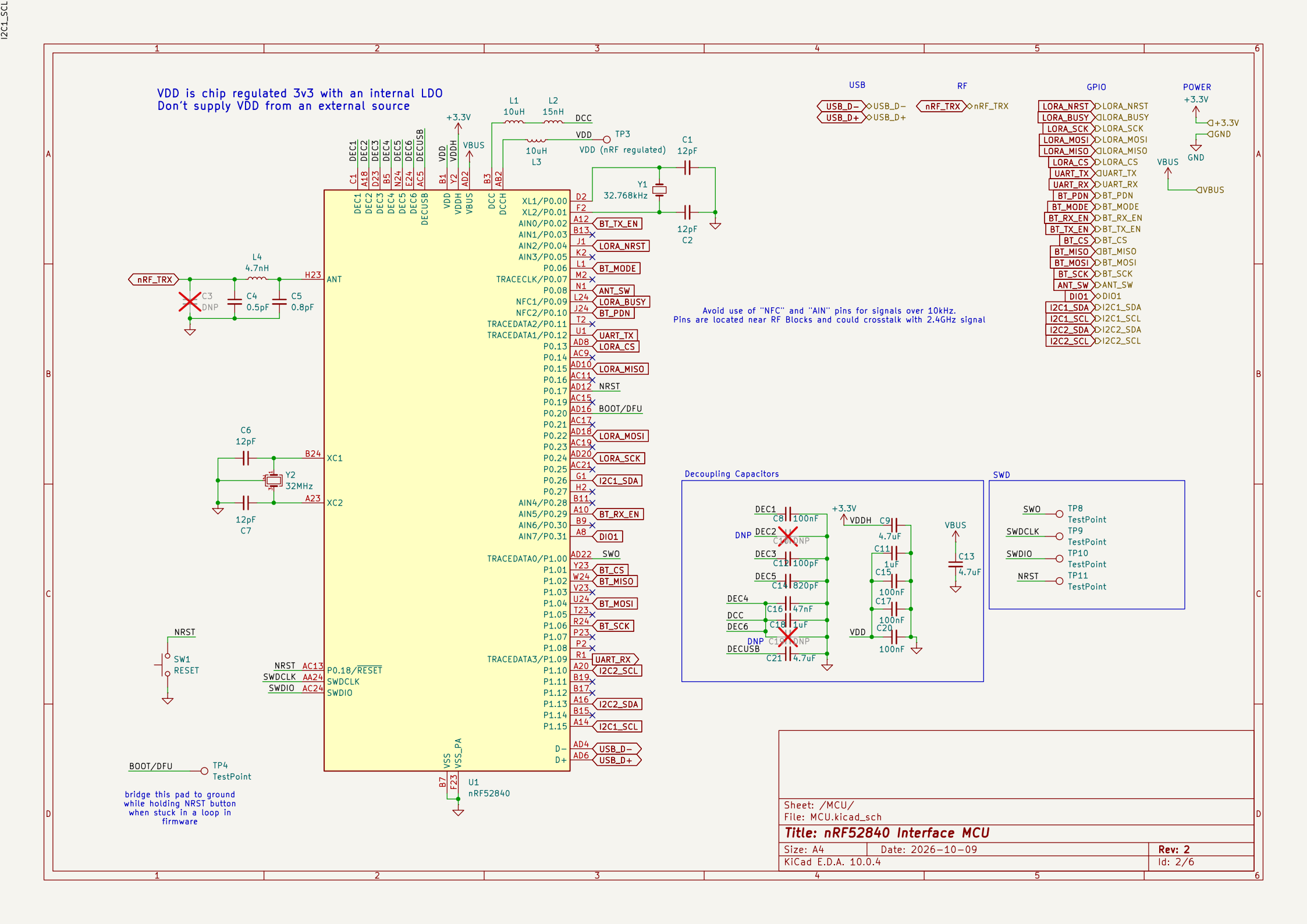
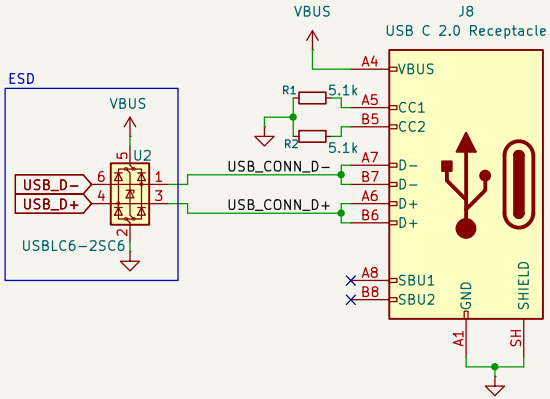
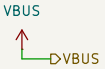

VIN is 2S
[6V,8.4V] (7V Nominal)
J1
Battery
VIN
GND
J2
UART
UART_TX
UART_RX
Power
+3.3V
GND
+5V
File: power.kicad_sch
MCU
BT_MODE
BT_RX_EN
BT_TX_EN
BT_PDN
BT_SCK
BT_MOSI
BT_MISO
BT_CS
nRF_TRX
Bluetooth
BT_TRX
File: bluetooth.kicad_sch
LoRa
LORA_BUSY
LORA_SCK
LORA_MOSI
LORA_MISO
LORA_CS
LORA_NRST
DIO1
ANT_SW
File: lora.kicad_sch
USB
USB_D+
USB_D−
VBUS
File: USB.kicad_sch
I2C1_SCL
I2C1_SDA
I2C2_SCL
I2C2_SDA
File: MCU.kicad_sch
5.1k for BOM consolidation,
standard is 4.7k for regular speed i2c
R19
R18
R20
R21
5.1k
J6
I2C 1
J3
I2C 2
J4
LoRa Test Conn
J5
BT Test Conn
TP1
TP2
H1
H2
H3
H4
MountingHole_Pad
Dylan DiBeneditto
Sheet: /
File: pcb.kicad_sch
Title: 2.4GHz BT and 915MHz LoRa Transciever
Size: A4
Date: 2026-10-09
Rev: 2
KiCad E.D.A. 10.0.4
Id: 1/6


VDD is chip regulated 3v3 with an internal LDO
Don't supply VDD from an external source
USB
RF
GPIO
POWER
Avoid use of "NFC" and "AIN" pins for signals over 10kHz.
Pins are located near RF Blocks and could crosstalk with 2.4GHz signal
Decoupling Capacitors
SWD
bridge this pad to ground while holding NRST button when stuck in a loop in firmware
U1
nRF52840
Sheet: /MCU/
File: MCU.kicad_sch
Title: nRF52840 Interface MCU
Size: A4
Date: 2026-10-09
Rev: 2
KiCad E.D.A. 10.0.4
Id: 2/6





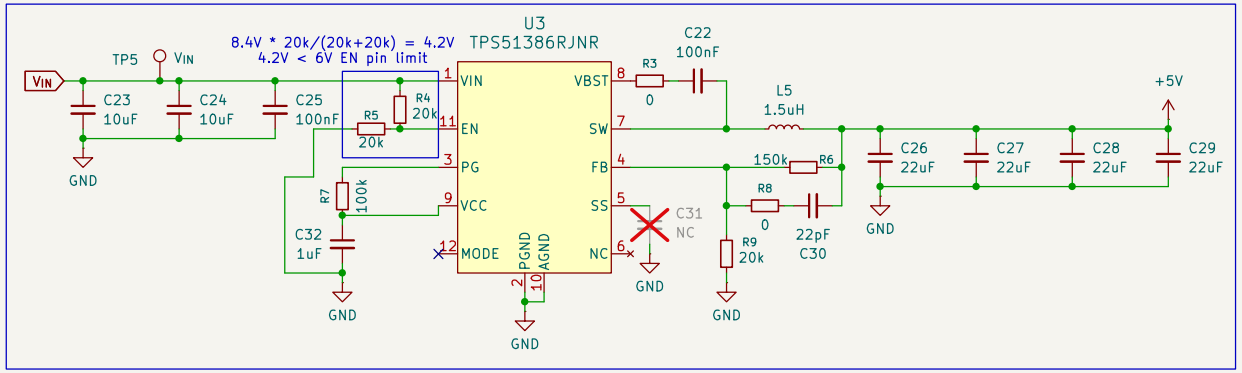


Dylan DiBeneditto		
Sheet: /USB/ File: USB.kicad_sch		
Title: USB−C 2.0 and ESD		
Size: A4	Date: 2026−10−09	**Rev: 2**
KiCad E.D.A. 10.0.4		Id: 3/6

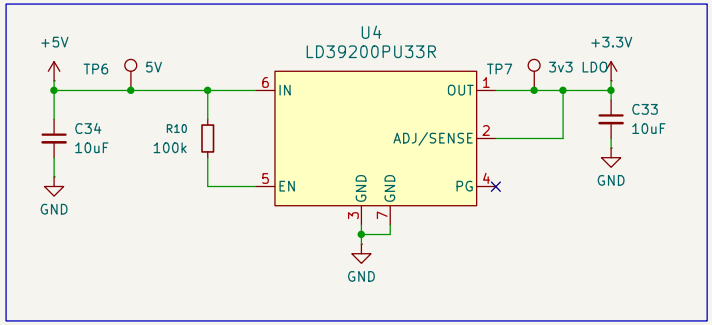
VIN is 2S

[6V,8.4V] (7V Nominal)

2S to 5V Buck Converter

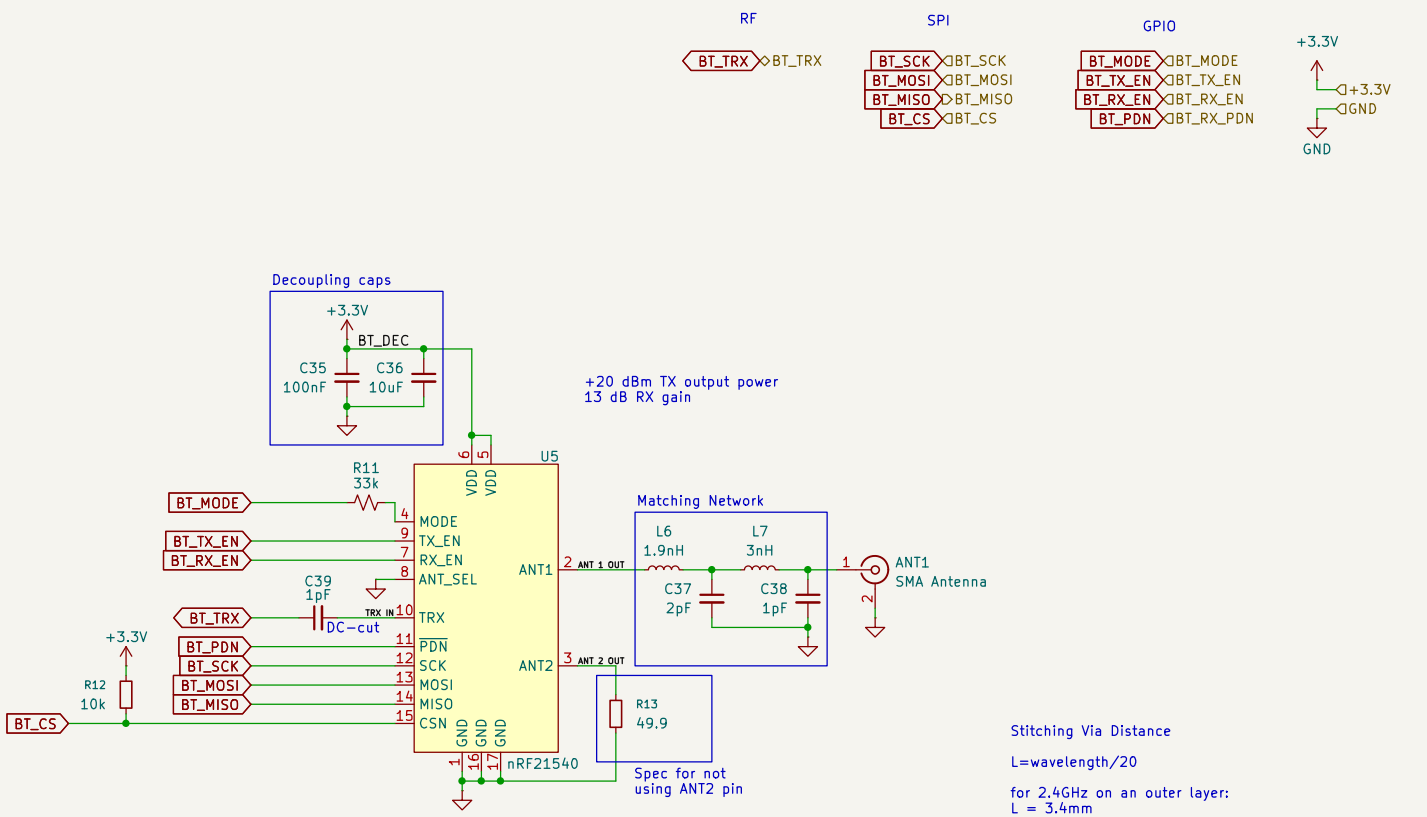


5V to 3.3V LDO



Dylan DiBeneditto	
Sheet: /Power/	
File: power.kicad_sch	
Title: Power	
Size: A4	Date: 2026-10-09
KiCad E.D.A. 10.0.4	Id: 4/6

Rev: 2



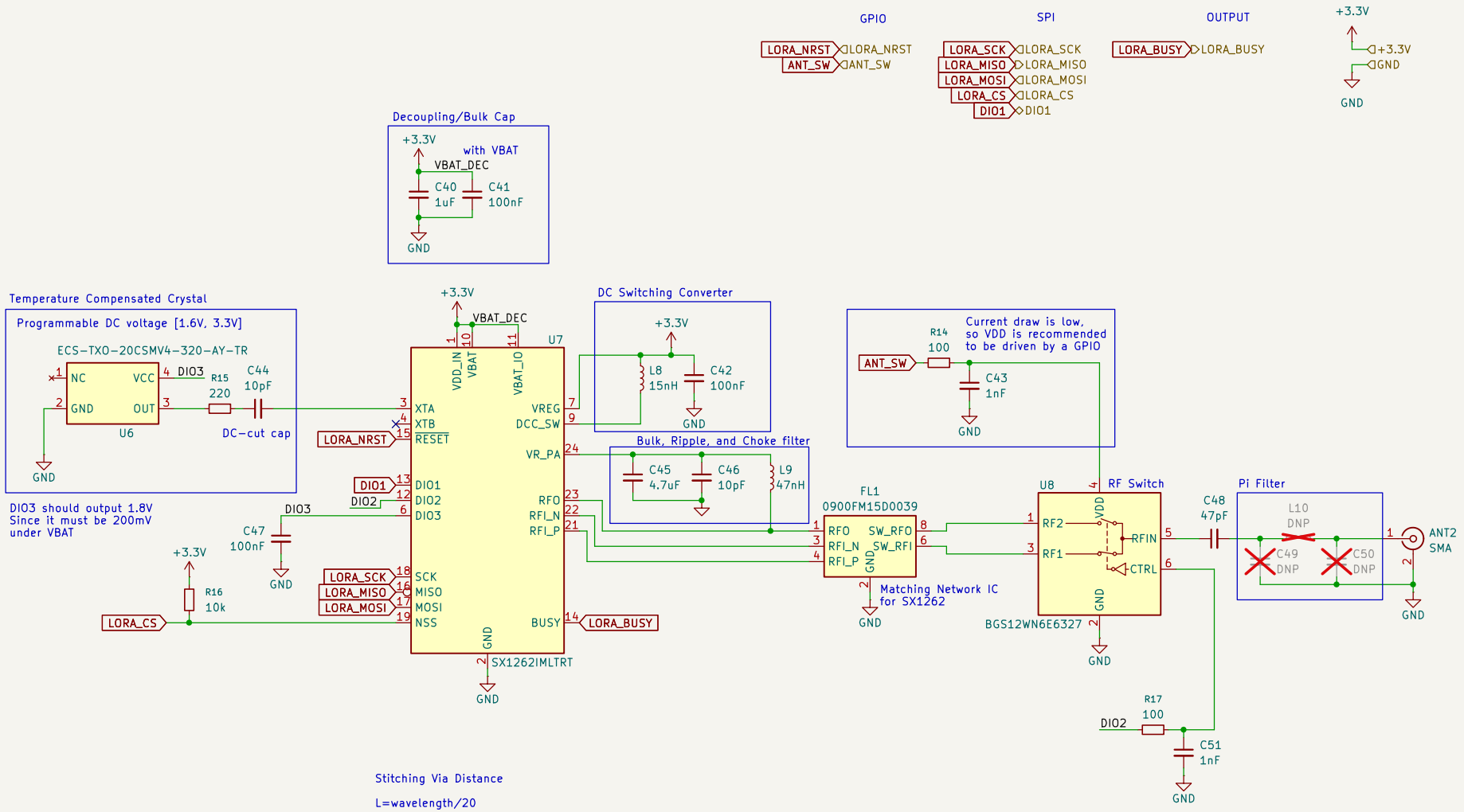
Stitching Via Distance

L=wavelength/20

for 2.4GHz on an outer layer:
L = 3.4mm

CPWG:
0.37mm Trace Width
0.22mm Gap
~49.9 ohm
(D_K = 4.2 with 7628 prepreg)
7628 prepreg thickness is 0.2104 according to JLCPCB

Dylan DiBeneditto		
Sheet: /Bluetooth/		
File: bluetooth.kicad_sch		
Title: Bluetooth		
Size: A4	Date: 2026-10-09	**Rev: 2**
KiCad E.D.A. 10.0.4		Id: 5/6



Dylan DiBeneditto

Sheet: /LoRa/
File: lora.kicad_sch

Title: 915MHz LoRa Modem

Size: A4 | Date: 2026-10-09 | Rev: 2
KiCad E.D.A. 10.0.4 | Id: 6/6